

Evaluation Metrics in Regression Models

1. Mean Absolute Error (MAE)

What MAE Measures:

Mean Absolute Error measures the average absolute difference between the actual values and the predicted values made by a regression model.

Mathematical Formula:

$$\text{MAE} = (1/n) \sum |y_i - \hat{y}_i|$$

Simple Mathematical Intuition:

1. Find the error for each prediction
2. Take the absolute value
3. Compute the average of all errors

How MAE Evaluates Regression Performance:

- Smaller MAE = better predictions
- Larger MAE = worse predictions
- MAE = 0 means perfect prediction

Advantages:

- Easy to understand
- Uses same unit as target variable
- Less sensitive to outliers

Disadvantages:

- Does not heavily punish large errors

Good vs Poor MAE Values:

Lower MAE indicates better model performance.

2. Mean Squared Error (MSE)

What MSE Measures:

Mean Squared Error measures the average squared difference between actual and predicted values.

Mathematical Formula:

$$\text{MSE} = (1/n) \sum (y_i - \hat{y}_i)^2$$

Simple Mathematical Intuition:

1. Calculate prediction errors
2. Square each error
3. Average all squared errors

How MSE Evaluates Regression Performance:

- Smaller MSE = better model
- Larger MSE = worse model

Advantages:

- Strongly penalizes large errors
- Widely used in machine learning

Disadvantages:

- Sensitive to outliers
- Units become squared

Good vs Poor MSE Values:

Lower MSE indicates better performance.

3. R² Score (Coefficient of Determination)

What R² Measures:

R² measures how well the regression model explains the variation in the data.

Mathematical Formula:

$$R^2 = 1 - [\Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2]$$

Simple Mathematical Intuition:

R² compares model error with the error from predicting the mean.

How R² Evaluates Regression Performance:

- R² = 1 means perfect prediction
- R² = 0 means model predicts like average
- R² < 0 means very poor model

Interpretation:

- 0.9+ = Excellent
- 0.7-0.9 = Good
- 0.5-0.7 = Moderate

- Below 0.5 = Weak

Advantages:

- Easy to compare models
- Measures explained variance

Disadvantages:

- Does not directly measure prediction error

Comparison of MAE, MSE, and R^2

MAE:

- Average absolute error
- Less sensitive to outliers

MSE:

- Average squared error
- Strongly penalizes large errors

R^2 :

- Measures explained variance
- Higher values indicate better fit

Final Understanding:

A good regression model usually has:

- Low MAE
- Low MSE

- High R^2 score
